

Stage 4 — Is it location, or which images carry the artifact?

Where the Shortcut Sits: progress report 4

In one sentence. The images whose artifact lies inside the ROI differ strongly from those whose artifact lies outside it (standardised differences up to 2.574), but the real-artifact crossover survives covariate matching and doubly robust regression adjustment in all three cohorts where matching is feasible (3 of 3); pasting the same real artifact inside or outside the ROI of the same images reproduces the location effect, although a neutral paste of the same shape reproduces most of it, so only a smaller part — significant for hair and thyroid calipers — is attributable to the artifact’s content; the crossover is also robust to the artifact-label definition, to stricter leakage groups and to a full rebuild of the data on a second machine.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma, malignant nodule, tumour with a solid component, erosion.
$A \in \{0, 1\}$	artifact present: a real artifact: hair, sonographer caliper, capsule debris — in its natural position (real traps) or pasted (transplant).
ROI	region of interest: lesion, nodule, tumour, union of expert lesion boxes; drawn in green; artifacts in red.
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	real traps: $[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$; transplant: location interaction $[\text{mask} - \text{ERM}]_{\text{pasted outside}} - [\text{mask} - \text{ERM}]_{\text{pasted inside}}$ on the same images (reversed-test AUROC).
CI	95% crossed seed×image bootstrap interval (Stage 1); every interval in this report uses it.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.
SMD	standardised mean difference of a covariate between two groups of images ($ \text{SMD} > 0.1$: imbalanced).

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1 Where this stage starts

Stage 3 showed a positive crossover with real artifacts in four cohorts and three encoder families. Its design compares two different sets of images: those whose artifact happens to lie inside the ROI (Trap A) and those whose artifact lies outside it (Trap B). Real artifacts are not placed at random. In dermoscopy, hair lies on large lesions more often than on small ones; in thyroid ultrasound, calipers inside the nodule are drawn on larger, more heavily measured nodules. Artifact-debiasing studies in dermoscopy make the same point (Bissoto et al., 2020; Nauta et al., 2022). The crossover could therefore reflect *which images* carry in-ROI artifacts rather than *where* the artifact lies. This stage separates the two, and then tests how fragile the result is to the choices that define it.

2 Questions

- Q4a** Do Trap-A and Trap-B images differ, and is the crossover explained by measured differences between them?
- Q4b** Does the location effect survive when the artifact is moved within the *same* image — and how much of that is due to the artifact rather than to pasting something into the image?
- Q4c** Is the crossover robust to how the artifact label is defined, to stricter leakage groups, and to rebuilding the whole pipeline?

Why three causal checks Matching removes measured imbalance but depends on the chosen covariates; regression on the matched sample (a doubly robust combination) protects against residual imbalance and against misspecification of either step alone; transplantation removes image-level confounding altogether but introduces its own artifact — the paste. The neutral paste (same mask shape, same compositing, filled with artifact-free tissue from the donor) measures that paste effect. *Rejected alternatives:* inverse-probability weighting (standardised differences above 2 in dermoscopy and capsule give extreme weights) and synthetic control images (they would repeat Stage 2, not test real artifacts).

3 Methods

Balance and matching (R1) Covariates fixed before any matching: dermoscopy — lesion area, hair amount, age, sex, body site, source, diagnosis within label; thyroid — nodule area, caliper pixels (log), native width and height (scanner proxy), mean grey level (gain proxy), nodule depth; ovary — tumour area, caliper pixels, image size, grey level, subtype within label; capsule — lesion-box area, contamination fraction, brightness, frame block (acquisition proxy). Standardised mean differences between the Trap-A and Trap-B artifact images, within label and pooled. Greedy 1:1 nearest-neighbour matching on the logit of a propensity model within label (exactly within source for dermoscopy), without replacement, calliper 0.2 SD; 0.05 SD as a stricter sensitivity analysis (R7). A cohort with fewer than 30 matched images per label is infeasible. The matched images replace the artifact cells of the traps; everything else is unchanged.

Regression adjustment (A2) For each reversed-test image of the matched sample, the *placement value* of the masked minus the ERM score (for a positive: the weighted share of negatives ranked below it; for a negative: the complement of the share of positives ranked above it), so that the class means reproduce the AUROC and the unadjusted difference between traps equals the crossover exactly (verified per seed). Within each seed, weighted least squares of this outcome on an intercept, the Trap-B indicator, the label and all matching covariates; the adjusted crossover is the Trap-B coefficient averaged over seeds. Crossed bootstrap with placement values and regression recomputed in every replicate (2,000 replicates); Holm over the three cohorts.

Transplant (R2) and neutral paste (R5) For each artifact-free recipient image, a real artifact instance is cut from a donor image of the same cohort (the donor’s mask pixels, largest connected component) and alpha-composited at a position fully inside ($\geq 95\%$ overlap) and at a position fully outside (0%) the recipient’s ROI, with a random dihedral transform; the positions are restricted to the field of view. The environments are then built as in the controlled sweeps with the pasted instance as the artifact; 5 presence seeds $\times$ 5 grouped folds. The neutral paste keeps the mask shape, transform, positions and compositing but fills the mask with tissue from the donor image at a window that does not touch its artifact.

Robustness (i) Artifact-label definitions: large detections only (≥ 50 caliper pixels) and strict location ($r \geq 0.7$ / $r < 0.05$) for thyroid and ovary; for capsule, a Trap B that also requires debris to cover less than 5% of the lesion box. (ii) Leakage: embedding-based near-duplicate groups (below). (iii) Rebuild: every dataset re-downloaded and every derived object — lesion U-Net, caliper detector, debris probe, leakage groups, feature caches — rebuilt on a second machine.

4 Experiments

Table 1: Experiments of this stage and their registration documents (docs/PREREGISTRATION_<name>.md or docs/<name>.md). Commit times are set by the committing machine and are not independent proof of order; the REVIEW2, REVIEW3 and FINAL registrations were also pushed to GitHub before their runs.

Experiment	Status	Registration	First commit (local time)	Support criterion	
Covariate balance (SMD) and matched traps, M1	supported registered	REVIEW2 (R1)	29cd0da, 18:56	2026-09-27	matched crossover CI > 0, Holm over 3 feasible cohorts
Stricter caliper 0.05 SD (R7)	supported registered	REVIEW3	85da735, 23:43	2026-09-27	sensitivity, descriptive
Real-artifact transplant, T1	supported registered	REVIEW2 (R2)	29cd0da, 18:56	2026-09-27	interaction CI > 0, Holm over 4
Neutral paste (paste-edge control), N3	supported registered	REVIEW3 (R5)	85da735, 23:43	2026-09-27	artifact — neutral interaction CI > 0
Regression-adjusted crossover, A2	supported registered	FINAL (A2)	c45d482, 00:35	2026-09-28	adjusted crossover CI > 0, Holm over 3
Artifact-label sensitivity	supported registered	SENSITIVITY_ARTIFACT_LABELS	c1bdfdf, 09:46	2026-09-27	crossover keeps sign and CI > 0 in every variant
Embedding-based leakage groups	supported registered	EMBEDDING_GROUPS	b0eda23, 13:55	2026-09-27	same sign and CI verdict as with pHash groups
Rebuild on a second machine, R0	supported registered	REVIEW2 (R0)	29cd0da, 18:56	2026-09-27	same sign, CI > 0, $ \Delta \leq 0.03$
Capsule matched analysis	not supported descriptive	REVIEW2 (R1)	29cd0da, 18:56	2026-09-27	infeasible under the registered rule

5 Results

5.1 The traps contain different images

The in-ROI and out-of-ROI artifact images differ strongly (Table 2): in dermoscopy the largest SMD is 2.127 (lesion area: hair lies inside large lesions), in capsule 2.574 (roi area), in thyroid 0.527 (caliper px log) and in ovary 0.389. Stage 3’s crossover is therefore not a comparison of like with like.

Table 2: Balance of pre-specified covariates between the in-ROI (Trap A) and out-of-ROI (Trap B) artifact images, before and after 1:1 nearest-neighbour matching within label (calliper 0.2 SD, pooled standardised mean differences). Registered rule: fewer than 30 matched images per label in either trap makes a cohort infeasible; target max $|\text{SMD}| \leq 0.1$ after matching.

Cohort	Least balanced covariate	max $ \text{SMD} $ before	after matching	min matched pairs per label	feasible (rule)
ISIC hair	lesion_area	2.127	0.108	398	yes
Thyroid	caliper_px_log	0.527	0.045	93	yes
Ovary	caliper_px_log	0.389	0.079	86	yes
Capsule	roi_area	2.574	0.231	19	no

Table 3: Per-covariate balance, thyroid and ovary (pooled over labels). The dermoscopy and capsule tables (17 and 20 covariates) are in `results/rerun.2026-09-28/review2/R1.balance.csv`.

Cohort	Covariate	SMD before	SMD after
Thyroid	roi_area	+0.395	<i>n/a</i>
Thyroid	caliper_px_log	+0.527	<i>n/a</i>
Thyroid	width	−0.143	<i>n/a</i>
Thyroid	height	−0.123	<i>n/a</i>
Thyroid	mean_grey	+0.183	<i>n/a</i>
Thyroid	roi_centroid_row	+0.060	<i>n/a</i>
Ovary	roi_area	+0.320	<i>n/a</i>
Ovary	caliper_px_log	+0.389	<i>n/a</i>
Ovary	width	−0.298	<i>n/a</i>
Ovary	height	+0.047	<i>n/a</i>
Ovary	mean_grey	+0.107	<i>n/a</i>
Ovary	subtype=0.0	+0.149	−0.079
Ovary	subtype=1.0	+0.059	+0.072
Ovary	subtype=2.0	−0.159	−0.012
Ovary	subtype=3.0	−0.054	+0.000
Ovary	subtype=4.0	−0.022	+0.024
Ovary	subtype=6.0	+0.024	+0.056
Ovary	subtype=7.0	−0.020	−0.062

5.2 Matching

Matching brings the maximum pooled SMD to 0.108 (dermoscopy), 0.045 (thyroid) and 0.079 (ovary); capsule matching is infeasible (19 pairs in the smaller label; SMD after matching 0.231). The crossover is essentially unchanged (Table 4): dermoscopy +0.154 [+0.123, +0.186], thyroid +0.238 [+0.186, +0.290], ovary +0.163 [+0.050, +0.273] (M1 supported in all three after Holm); with the stricter calliper it shrinks by 9–16% and survives: +0.124 [+0.086, +0.162], +0.218 [+0.165, +0.270], +0.152 [+0.041, +0.265]. Capsule, matched descriptively, gives +0.230 [+0.036, +0.434] — lower than unmatched (+0.377 [+0.327, +0.426]) and imprecise, so capsule’s real-trap crossover is reported as descriptive.

Table 4: Crossover before and after covariate matching (reversed-test AUROC, crossed 95% CIs). The 0.05 SD calliper is a stricter, registered sensitivity analysis (R7).

Cohort	Crossover, all images	matched (0.2 SD)	matched (0.05 SD)	M1 (Holm, 3)
ISIC hair	+0.147 [+0.111, +0.185]	+0.154 [+0.123, +0.186]	+0.124 [+0.086, +0.162]	supported
Thyroid	+0.239 [+0.196, +0.280]	+0.238 [+0.186, +0.290]	+0.218 [+0.165, +0.270]	supported
Ovary	+0.170 [+0.087, +0.252]	+0.163 [+0.050, +0.273]	+0.152 [+0.041, +0.265]	supported
Capsule	+0.377 [+0.327, +0.426]	+0.230 [+0.036, +0.434]	+0.328 [+0.087, +0.573]	descriptive (infeasible)

5.3 Regression adjustment on the matched sample (A2)

Within-label imbalance up to about 0.16 remained after matching. The doubly robust adjusted crossover equals the unadjusted one to the third decimal in every cohort (Table 5): dermoscopy

+0.156 [+0.127, +0.187] (unadjusted +0.154 [+0.124, +0.186]), thyroid +0.238 [+0.184, +0.291] (+0.238 [+0.185, +0.291]), ovary +0.165 [+0.056, +0.273] (+0.163 [+0.058, +0.268]); Holm-adjusted $p = 0.002, 0.002$ and 0.002 . All 3 registered tests are supported. Residual measured imbalance does not carry the effect. What this cannot exclude is an *unmeasured* difference.

Table 5: Regression-adjusted crossover on the matched traps (A2): per-image placement values of the masked minus the ERM score regressed on the Trap-B indicator, the label and all matching covariates, within seed; crossed bootstrap (2,000 replicates). Capsule is not analysed (matching infeasible; descriptive only).

Cohort	Test images per seed	Covariate columns	Post-matching max SMD pooled / per label	Unadjusted crossover	Adjusted (doubly robust)	p (Holm, 3)	A2
ISIC hair	2960	17	0.108 / 0.164	+0.154 [+0.124, +0.186]	+0.156 [+0.127, +0.187]	0.002	supported
Thyroid	2150	6	0.045 / 0.150	+0.238 [+0.185, +0.291]	+0.238 [+0.184, +0.291]	0.002	supported
Ovary	576	11	0.079 / 0.140	+0.163 [+0.058, +0.268]	+0.165 [+0.056, +0.273]	0.002	supported

Table 6: Post-matching balance table of the A2 sample (pooled over labels; test images of the reversed environment).

Cohort	Covariate	Trap A mean	Trap B mean	SMD
ISIC hair	lesion_area	0.297	0.292	+0.033
ISIC hair	hair_amount	0.018	0.015	+0.108
ISIC hair	age	57.119	57.332	-0.012
ISIC hair	age_missing	0.019	0.026	-0.048
ISIC hair	sex=female	0.504	0.521	-0.035
ISIC hair	sex=male	0.480	0.457	+0.046
ISIC hair	sex=missing	0.016	0.022	-0.040
ISIC hair	site=head/neck	0.174	0.179	-0.013
ISIC hair	site=lower extremity	0.219	0.213	+0.015
ISIC hair	site=missing	0.076	0.072	+0.015
ISIC hair	site=oral/genital	0.001	0.002	-0.015
ISIC hair	site=palms/soles	0.027	0.030	-0.020
ISIC hair	site=torso	0.383	0.387	-0.008
ISIC hair	site=upper extremity	0.120	0.118	+0.008
ISIC hair	benign_dx=BKL	0.092	0.100	-0.026
ISIC hair	benign_dx=MEL	0.218	0.218	+0.000
ISIC hair	benign_dx=Nv	0.442	0.413	+0.060
ISIC hair	benign_dx=other	0.248	0.270	-0.050
ISIC hair	source=BCN	0.545	0.545	+0.000
ISIC hair	source=HAM	0.386	0.386	+0.000
ISIC hair	source=MSK	0.069	0.069	+0.000
Thyroid	roi_area	0.117	0.118	-0.008
Thyroid	caliper_px_log	5.495	5.448	+0.037
Thyroid	width	450.288	441.291	+0.045
Thyroid	height	373.576	371.744	+0.017
Thyroid	mean_grey	60.457	60.609	-0.006
Thyroid	roi_centroid_row	0.367	0.366	+0.007
Ovary	roi_area	0.141	0.142	-0.001
Ovary	caliper_px_log	6.518	6.481	+0.032
Ovary	width	864.648	856.939	+0.037
Ovary	height	550.073	549.944	+0.003
Ovary	mean_grey	57.307	56.935	+0.022
Ovary	subtype=0.0	0.218	0.251	-0.079
Ovary	subtype=1.0	0.201	0.173	+0.072
Ovary	subtype=2.0	0.296	0.302	-0.012
Ovary	subtype=3.0	0.089	0.089	+0.000
Ovary	subtype=4.0	0.061	0.056	+0.024
Ovary	subtype=6.0	0.106	0.089	+0.056
Ovary	subtype=7.0	0.028	0.039	-0.062

5.4 The same artifact inside and outside the ROI

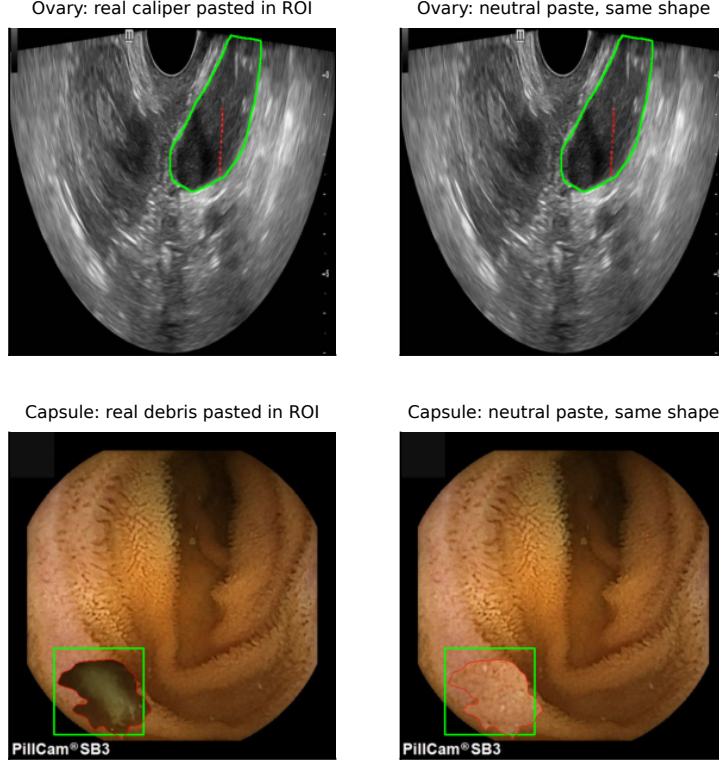


Figure 1: Transplant and neutral paste on the same recipient images (openly licensed cohorts; MMOTU, SEE-AI CC BY 4.0). Left: a real artifact instance pasted inside the ROI (green; the pasted region outlined in red). Right: the neutral paste — same mask shape and compositing, filled with artifact-free tissue from the donor.

Pasting the same real instance outside versus inside the ROI yields a positive location interaction in every cohort (Table 7; dermoscopy $+0.253$ [$+0.214, +0.293$], thyroid $+0.277$ [$+0.256, +0.298$], ovary $+0.074$ [$+0.044, +0.104$], capsule $+0.497$ [$+0.427, +0.567$]; Holm over four). Table 8 shows the mechanism of Stage 2 on real artifacts: outside the ROI masking removes the pasted artifact completely — the masked model’s correlated and reversed AUROC coincide (e.g. thyroid 0.846 and 0.846) and $|\Delta p| = 0$ — whereas inside it the artifact survives, and for thyroid, ovary and capsule the masked model reacts to it *more* than ERM (capsule 0.551 versus 0.238). Group balancing has a small location interaction of the opposite sign (e.g. -0.052 [$-0.071, -0.033$] for hair), much smaller than masking’s: it is nearly location-invariant.

Table 7: Real-artifact transplant (T1): the same real artifact instance, cut from a donor image, pasted outside and inside the ROI of the same artifact-free recipient images. These effects combine the artifact with the paste itself (seam, occluded tissue); Table 9 separates them.

Cohort	mask – ERM, pasted outside ROI	pasted inside ROI	Location interaction	Balanced: interaction
ISIC hair	$+0.178$ [$+0.128, +0.227$]	-0.075 [$-0.114, -0.036$]	$+0.253$ [$+0.214, +0.293$]	-0.052 [$-0.071, -0.033$]
Thyroid	$+0.349$ [$+0.321, +0.379$]	$+0.072$ [$+0.043, +0.100$]	$+0.277$ [$+0.256, +0.298$]	-0.016 [$-0.025, -0.007$]
Ovary	$+0.064$ [$+0.001, +0.126$]	-0.010 [$-0.075, +0.056$]	$+0.074$ [$+0.044, +0.104$]	-0.011 [$-0.025, +0.002$]
Capsule	$+0.322$ [$+0.262, +0.384$]	-0.175 [$-0.241, -0.109$]	$+0.497$ [$+0.427, +0.567$]	-0.032 [$-0.068, +0.004$]

Table 8: Transplant, per location: correlated and reversed AUROC of ERM and of the masked model, and the same-head counterfactual sensitivity to the pasted artifact (T3). Outside the ROI the masked model’s correlated and reversed AUROC coincide and $|\Delta p| = 0$: the pasted artifact is gone.

Cohort	Recipients	ERM in: corr / rev	mask in: corr / rev	mask out: corr / rev	$ \Delta p $ in: ERM / mask	$ \Delta p $ out: mask
ISIC hair	860	0.904 / 0.454	0.842 / 0.379	0.716 / 0.716	0.340 / 0.297	0.000
Thyroid	1800	0.851 / 0.487	0.906 / 0.559	0.846 / 0.846	0.206 / 0.271	0.000
Ovary	348	0.771 / 0.657	0.825 / 0.647	0.750 / 0.750	0.037 / 0.116	0.000
Capsule	340	0.906 / 0.508	0.956 / 0.333	0.914 / 0.914	0.238 / 0.551	0.000

How much of it is the artifact? The neutral paste reproduces 52% to 92% of the interaction (Table 9). Pasting *anything* into the ROI changes what masking does, because the environments make the pasted region label-correlated whatever it contains: the neutral paste becomes a shortcut itself (ERM learns it), and masking keeps it only when it lies inside the ROI. Only the difference between the two — N3, the interaction with the real artifact minus the interaction with the neutral paste — is attributable to the artifact’s content: dermoscopy +0.049 [+0.026, +0.072], thyroid +0.132 [+0.116, +0.148], ovary +0.022 [−0.002, +0.048] and capsule +0.037 [−0.017, +0.093] (2 of 4 excluding zero). By the registered decision rule (N3 significant *and* neutral share below one quarter) the transplant signal is not attributable to the artifact alone in any cohort.

What the transplant does and does not show It shows that on *fixed images*, where a label-correlated pattern sits decides what masking removes: the image-population explanation of Q4a is excluded. It does *not* give the harm of a real artifact in its natural context: the effect of masking with pasted hair inside the lesion combines the hair with the paste and is not attributed to real hair. For hair specifically, the transplant constructs a pixel-carried shortcut, whereas natural hair is largely correlate-carried (Stage 3).

Table 9: Paste-edge control (R5): the same mask shape and compositing, filled with the donor’s own artifact-free tissue. Only the N3 column — the interaction with the real artifact minus the interaction with the neutral paste — is attributable to the artifact’s content.

Cohort	mask − ERM, neutral outside	neutral inside	Neutral interaction	Artifact − neutral (N3)	Neutral / artifact interaction
ISIC hair	+0.129 [+0.079, +0.181]	−0.075 [−0.116, −0.035]	+0.204 [+0.164, +0.244]	+0.049 [+0.026, +0.072]	81%
Thyroid	+0.200 [+0.175, +0.226]	+0.055 [+0.028, +0.081]	+0.145 [+0.130, +0.161]	+0.132 [+0.116, +0.148]	52%
Ovary	+0.072 [+0.009, +0.135]	+0.019 [−0.041, +0.080]	+0.052 [+0.021, +0.084]	+0.022 [−0.002, +0.048]	71%
Capsule	+0.215 [+0.154, +0.280]	−0.244 [−0.314, −0.174]	+0.459 [+0.383, +0.535]	+0.037 [−0.017, +0.093]	92%

5.5 Robustness of the crossover

Artifact-label definition Table 10. With only large caliper detections, which are almost certainly real calipers, the crossover is +0.244 [+0.199, +0.289] (thyroid) and +0.188 [+0.108, +0.266] (ovary); with strict location thresholds +0.212 [+0.164, +0.259] and +0.196 [+0.117, +0.277]. It keeps its sign and significance in every variant and grows for the ovary, as expected if detector errors dilute rather than create the effect. For capsule, a Trap B that also excludes debris covering part of the lesion box gives +0.255 [+0.153, +0.349] — smaller than the main +0.377 [+0.327, +0.426] (the strict Trap B has far fewer benign images) but clearly positive; the expectation that the main definition was conservative was not confirmed.

Table 10: Sensitivity of the crossover to the definition of the artifact label (registered before running): only large detections, stricter location thresholds, and for capsule a Trap B that also requires debris to cover less than 5% of the lesion box.

Cohort	Artifact-label definition	Artifact images A / B	Crossover
Thyroid	main	1150 / 326	+0.239 [+0.196, +0.280]
Thyroid	large markers (≥ 50 px)	1083 / 274	+0.244 [+0.199, +0.289]
Thyroid	strict location ($r \geq 0.7 / r < 0.05$)	919 / 298	+0.212 [+0.164, +0.259]
Ovary	main	617 / 194	+0.170 [+0.087, +0.252]
Ovary	large markers (≥ 50 px)	609 / 183	+0.188 [+0.108, +0.266]
Ovary	strict location ($r \geq 0.7 / r < 0.05$)	571 / 186	+0.196 [+0.117, +0.277]
Capsule	main	816 / 1240	+0.377 [+0.327, +0.426]
Capsule	strict Trap B (lesion coverage $< 5\%$)	816 / 373	+0.255 [+0.153, +0.349]

Leakage without patient identifiers TN3K, MMOTU and SEE-AI publish no patient or video identifiers, so the main analysis groups near-duplicates by perceptual hash (and, for capsule, frame blocks). A stricter grouping joins images whose DINOv2 embeddings are near-identical (Table 11). Raw embedding cosine was dominated by the shared ultrasound appearance (median nearest-neighbour similarity 0.99 in the ovary), so embeddings were mean-centred per cohort — a change recorded before any model was refitted. The crossover is unchanged: thyroid +0.235 [+0.193, +0.275], ovary +0.157 [+0.053, +0.255], capsule +0.353 [+0.296, +0.410]. Embedding groups remain a proxy for patient identity: they remove visually near-identical images, not different-looking images of one patient.

Table 11: Stricter leakage groups for the cohorts without patient identifiers: images joined whenever their (mean-centred) DINOv2 embeddings have cosine $\geq \tau$, united with the existing groups; τ is the most aggressive grid value that keeps the largest group below 5% of the cohort (chosen without labels).

Cohort	Leakage groups	cosine τ	largest group	Crossover, pHash groups	Crossover, embedding groups
Thyroid	3,455 $\rightarrow$ 3,081	0.80	4.6%	+0.239 [+0.196, +0.280]	+0.235 [+0.193, +0.275]
Ovary	1,059 $\rightarrow$ 969	0.80	2.1%	+0.170 [+0.087, +0.252]	+0.157 [+0.053, +0.255]
Capsule	137 $\rightarrow$ 135	0.93	3.7%	+0.377 [+0.327, +0.426]	+0.353 [+0.296, +0.410]

Rebuild on a second machine Table 12. After every dataset was downloaded again and every derived object rebuilt on different hardware, each crossover reproduces within 0.01 (registered tolerance 0.03). The re-trained lesion U-Net changed hair-trap membership by about 3%, and the retrained debris probe moved a handful of frames across the 3%/10% thresholds; neither matters. All intervals in Stages 3–4 come from the rebuilt data.

Table 12: Reproduction of the four crossovers after every dataset was re-downloaded and every derived object (lesion U-Net, detectors, probe, groups, caches) rebuilt on a second machine (registered criterion: same sign, CI excluding 0, $|\Delta| \leq 0.03$).

Cohort	First build (point)	Rebuilt data, second machine	$ \Delta $
ISIC hair	+0.145	+0.147 [+0.111, +0.185]	0.003
Thyroid	+0.230	+0.239 [+0.196, +0.280]	0.009
Ovary	+0.170	+0.170 [+0.087, +0.252]	0.001
Capsule	+0.368	+0.377 [+0.327, +0.426]	0.008

5.6 Sensitivity to the interval estimator

On the analyses of this stage (Table 13, Figure 2) the crossed intervals are wider by a median factor of 1.63; 3 of 204 contrasts lose significance (Table 14), none of them a crossover, matched crossover

or transplant interaction of masking.

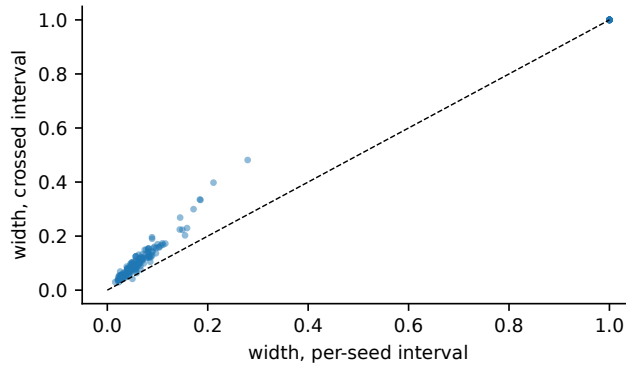


Figure 2: Width of each crossed interval against the per-seed interval of the same contrast (dashed: equal width).

Table 13: The analyses of this stage under the per-seed and the crossed estimator (same predictions; baseline arms only).

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the estimator	0 only crossed	Median width ratio (crossed / per-seed)
capsule/dino518_emb_groups	9	0		0		2.02
capsule/dino518_matched	5	0		0		1.75
capsule/dino518_matched05	5	0		0		1.72
capsule/dino518_repro	5	0		0		1.50
capsule/dino518_sens_strictB	9	0		0		1.66
ovary/dino518_emb_groups	9	0		0		1.55
ovary/dino518_matched	5	0		0		1.51
ovary/dino518_matched05	5	0		0		1.52
ovary/dino518_repro	5	0		0		1.55
ovary/dino518_sens_px50	9	0		0		1.62
ovary/dino518_sens_strictloc	9	0		0		1.59
review2/transplant	60	1		0		1.63
spec_e13/dino518_matched	9	0		0		1.65
spec_e13/dino518_matched05	9	1		0		1.63
spec_e13/dino518_repro	9	0		0		1.71
thyroid/dino518_emb_groups	9	0		0		1.56
thyroid/dino518_matched	5	0		0		1.67
thyroid/dino518_matched05	5	0		0		1.69
thyroid/dino518_repro	5	0		0		1.76
thyroid/dino518_sens_px50	9	0		0		1.58
thyroid/dino518_sens_strictloc	9	1		0		1.63
all	204	3		0		1.63

Table 14: Contrasts of this stage whose verdict depends on the estimator.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
thyroid/dino518_sens_strictloc	trapB / balanced / clean / balanced / erm	+0.043 [−0.008, +0.091]	no longer excludes 0
spec_e13/dino518_matched05	trapA / mask / clean / all / mask / erm	−0.026 [−0.053, +0.000]	no longer excludes 0
review2/transplant/ ovary_dino518	0.0 / balanced / erm / test_rev	+0.019 [−0.006, +0.046]	no longer excludes 0

6 What failed or changed

- **The transplant is mostly a paste effect.** The neutral paste reproduces 52–92% of the interaction; by the registered rule the transplant is not attributable to the artifact alone in any

cohort. No harm figure from the transplant is attributed to real artifacts.

- **Capsule cannot be matched** under the registered rule; its real-trap crossover is descriptive, and its location evidence rests on the transplant.
- **Stricter matching does not improve within-label balance** (up to 0.18–0.19) and shrinks the crossover by 9–16%.
- **The capsule strict-Trap-B variant** gives a smaller crossover, contrary to the expectation registered with it.

7 Anticipated questions

Why not simply adjust for lesion size in the original traps? Adjustment without matching extrapolates where the traps do not overlap (hair SMD above 2); matching first restricts the comparison to images that exist in both traps, and regression then removes what imbalance remains. Doing both is doubly robust: either step alone being correct is enough.

Why placement values? AUROC is not a sum over images, so it cannot be regressed directly. Placement values decompose it exactly into per-image contributions whose class means reproduce the AUROC, so a regression on them adjusts the crossover itself.

Why is capsule infeasible? In-ROI debris goes with large lesion boxes (SMD 2.6); among polyp-like lesions only 19 matched pairs exist, below the registered minimum of 30.

If a neutral paste gives most of the effect, is the transplant useless? No. It answers the question it was designed for — whether the location effect exists on fixed images — and the neutral paste answers a second question: the effect is about *any* label-correlated pattern inside the ROI, of which a real artifact is one example. What it cannot give is an estimate of harm for a particular real artifact.

Why does the neutral paste become a shortcut? Because the environments make its presence label-correlated, and a pasted patch leaves a visible seam and occludes the tissue beneath; the encoder can detect both.

Could unmeasured confounding still explain the real-trap crossover? In principle, yes; matching and regression only handle measured covariates. The transplant shows that location alone produces the effect on fixed images, which makes an unmeasured-confounder explanation unnecessary, but it cannot rule one out for the natural traps.

Why group by embeddings rather than ask for patient identifiers? The data providers do not publish them. Embedding groups are the strictest available proxy; they reduced thyroid groups from 3,455 to 3,081 without changing the result.

Why rebuild everything instead of re-running the models? Re-running on cached objects tests only the last step. Rebuilding tests the whole chain — downloads, masks, detectors, groups, features — and found two gaps in the reproduction recipe, which were fixed.

8 Conclusion

Supported. The real-artifact crossover is not explained by measured covariates: it survives matching (3 of 3 feasible cohorts) and doubly robust adjustment (3 of 3, Holm). Moving the same real artifact inside the ROI of fixed images reproduces the location effect in all four cohorts, and for hair and thyroid calipers the artifact’s content adds to the effect of the paste. The crossover is robust to the artifact-label definition, to stricter leakage groups and to a full rebuild. **Qualified.** Most of the transplant interaction is produced by the paste itself; unmeasured confounding of the natural location cannot be excluded; capsule is descriptive. **Not supported.** A harm figure for real artifacts from the transplant; the capsule matched analysis.

9 Questions for the supervisor

1. Is matching plus doubly robust adjustment plus the transplant sufficient causal evidence for the thesis, or should a negative-control analysis be attempted as well?
2. How should the paste effect be presented: as a limitation of the transplant, or as a finding in itself (any label-correlated pattern inside the ROI defeats masking)?
3. Should capsule endoscopy remain a primary cohort given that its real-trap crossover is descriptive?
4. Is the embedding-group analysis an acceptable answer to the absence of patient identifiers, or must the limitation be stated more strongly?

References

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